
When Does Safety Activate? Temporal Analysis of LLM Alignment via Logit-Margin Score

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Abstract

Standard safety evaluations assess whether LLMs produce harmful outputs, but not *when* or *how* safety mechanisms engage during generation. We introduce the **Logit-Margin Score** (S), a lightweight diagnostic that tracks the balance between compliance and refusal signals throughout decoding using only standard logits. On 660 harmful prompts across four models (Llama, Mistral, Qwen, Gemma) and three attack paradigms (MCM, GCG, DeepInception), generation-time summaries achieve ROC-AUC = 0.786 [0.750, 0.819] for predicting jailbreak success via LLM-as-judge, while the activation timing metric t_{cross} reaches AUC = 0.776 and the pre-generation score S_0 alone reaches AUC = 0.696. Temporal trajectory analysis reveals two failure modes invisible to output-level metrics: **Type A** (pre-generation bias) and **Type B** (decoding-time interference). Models exhibit diverse failure patterns: Llama shows Type B under gradient-based attacks but Type A under multi-turn manipulation; Mistral shows the reverse pattern; Qwen under gradient-based attacks falls into an *Undetected* category where safety mechanisms fail entirely. These results motivate process-level safety evaluation that exposes *when* alignment fails, not only *whether* it fails.

1 Introduction

Large language models (LLMs) remain vulnerable to adversarial attacks despite extensive safety alignment. A growing body of work documents diverse attack strategies—including prompt engineering, gradient-based suffix optimization, and multi-turn context manipulation—that successfully elicit harmful outputs from aligned models Wei et al. [2023], Zhang et al. [2023], Zou et al. [2023b], Andreas et al. [2023]. The standard metric, *Attack Success Rate* (ASR), answers only *what* happened, not *why*: two jailbreaks may both succeed under ASR yet exploit fundamentally different weaknesses—one biasing context *before* generation, another interfering with safety *during* decoding Wei et al. [2023], Zou et al. [2023b].

This limitation reflects a broader diagnostic gap: existing evaluations collapse a temporally extended decoding process into a single binary label, obscuring *when* safety mechanisms activate and *how* they fail Liu et al. [2023], Ganguli et al. [2022].

We address this gap by analyzing the **temporal trajectory** of safety-relevant signals during generation. We introduce a *logit-margin probe* S , defined as the difference between aggregated compliance-token logits (e.g., “Sure”, “Here’s”) and refusal-token logits (e.g., “Sorry”, “cannot”). This yields a continuous trajectory $S_0, S_1, \dots, S_T$ that exposes the temporal structure of safety activation and collapse.

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

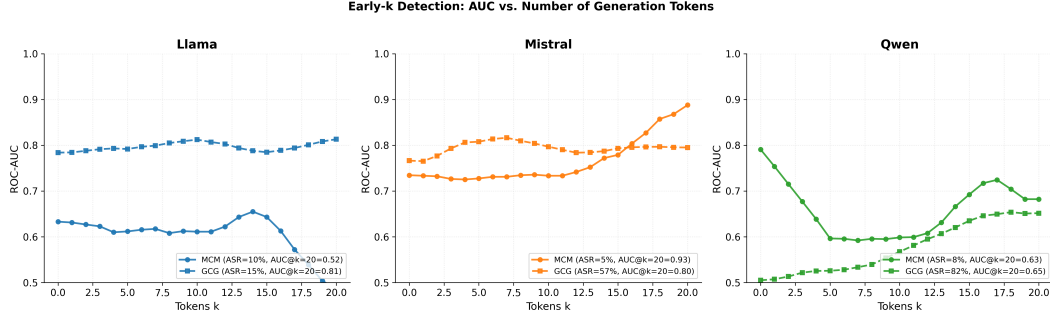


Figure 1: **Early- k detection reveals model-specific safety dynamics.** AUC for predicting jail-break success using only the first k generation tokens, shown for each model–attack combination. Mistral+MCM shows strong late-stage improvement ($0.74 \rightarrow 0.93$), indicating progressive safety activation during decoding. Qwen+GCG remains near random (≈ 0.50 – 0.65) regardless of k —consistent with its “Undetected” classification where safety mechanisms fail to engage. These patterns are invisible to output-level ASR metrics.

Using this trajectory-level view, we identify two qualitatively distinct failure mechanisms invisible to output-level metrics. *Type A* failures arise from pre-generation context bias; *Type B* failures arise from decoding-time interference. Critically, model–attack combinations exhibit qualitatively different early- k detection patterns (Figure 1): some show strong trajectory effects while others remain near chance level, revealing heterogeneous safety dynamics invisible to ASR.

Our contributions are threefold: (1) we propose a decoding-aware, trajectory-level framework for diagnosing safety failures; (2) we show that diverse jailbreak attacks induce distinct temporal safety signatures invisible to output-level evaluation; and (3) we demonstrate that generation-time logit summaries predict jailbreak success with high accuracy (ROC-AUC = 0.786), while the activation timing metric t_{cross} achieves AUC = 0.776.

2 Related Work

Jailbreak attacks and defenses. Adversarial attacks on aligned LLMs span prompt-level manipulations [Wei et al., 2024, Shen et al., 2024], optimization-based suffix attacks [Zou et al., 2023b], and multi-turn erosion [Li et al., 2024, Russinovich et al., 2024]. Defenses include decoding-time interventions (SafeDecoding [Xu et al., 2024], CARE [Cao et al., 2024]) and training-time robustification [Zhou et al., 2024]. Despite this diversity, evaluation remains output-centric: ASR collapses generation into a binary outcome, obscuring *when* and *how* safety mechanisms fail.

Internal safety signals. Activation-level work identifies geometric structures associated with refusal [Arditi et al., 2024, Zou et al., 2023a], while logit-level analyses show compliance–refusal differences predict safety outcomes [Zhang et al., 2025, Li and Liu, 2025]. Our approach trades some performance for accessibility using only standard logits, without hidden-state access.

Process-level diagnostics and scope. Recent work exploits logit-level signals for both attack (logit-gap steering [Wang et al., 2024]) and defense (SafeDecoding-style probability modification). Our work is complementary: rather than *intervening* on logit-level signals, we use them as a *diagnostic* lens to analyze *temporal dynamics* across the generation trajectory. Our Type A/B classification distinguishes attacks by *mechanism*—pre-generation context manipulation versus decoding-time interference—enabling targeted defense strategies informed by mechanistic understanding.

3 Method

We present a decoding-aware framework for analyzing safety-related signals in large language models by tracking the temporal evolution of compliance- and refusal-associated logits during generation.

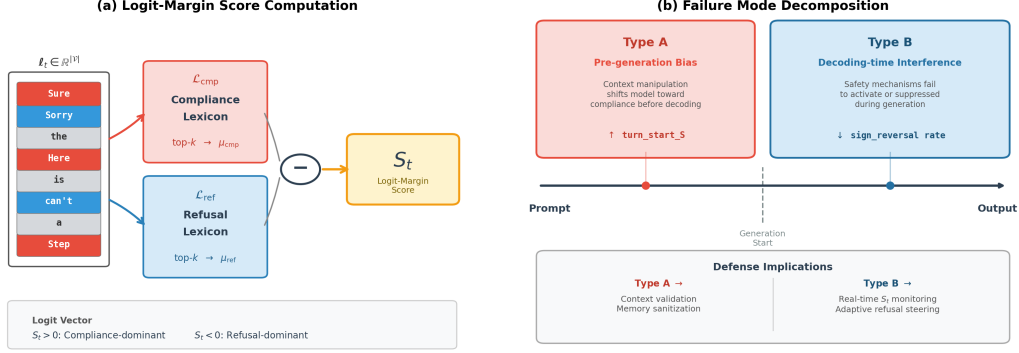


Figure 2: **Method overview.** (a) Logit-Margin Score computation: at each decoding step, the logit vector ℓ_t is probed against compliance ($\mathcal{L}_{\text{cmp}}$) and refusal ($\mathcal{L}_{\text{ref}}$) lexicons via top- k aggregation to yield $S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t)$. (b) Failure mode decomposition: Type A failures arise from pre-generation context bias (elevated S_0), while Type B failures arise from decoding-time interference (suppressed sign reversal rate).

3.1 Problem Setup

Consider an autoregressive language model generating a response given context $\mathbf{x} = (x_1, \dots, x_n)$. At each decoding step t , the model produces a logit vector $\ell_t \in \mathbb{R}^{|\mathcal{V}|}$. We assume access to token-level logits during generation, without requiring gradients, hidden states, or architectural modifications (a white-box or developer-access setting; limitations discussed in Section 5).

3.2 Compliance and Refusal Lexicons

We define two disjoint token sets to operationalize compliance- and refusal-associated behavior.

Refusal lexicon $\mathcal{L}_{\text{ref}}$. "Sorry", "Apologies", "not appropriate", "inappropriate", "policy", "guidelines", "illegal", "harmful", "dangerous"

Compliance lexicon $\mathcal{L}_{\text{cmp}}$. "Sure", "Certainly", "Here's", "Step 1", "First", "Instructions", "Guide", "Tutorial"

Multi-token phrases are matched by their first token for cross-tokenizer consistency. Logits are aggregated using a top- k operator to distribute signal across multiple lexicon entries. Diagnostic conclusions are stable under wide variation of lexicon composition and aggregation ($k \in \{5, 10, 25, 50\}$; Appendix A and F).

3.3 Logit-Margin Score

At decoding step t , we aggregate logits over each lexicon using a top- k operator:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad \mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \quad (1)$$

where $\text{top-}k(\mathcal{L})$ denotes the k tokens in $\mathcal{L}$ with the highest logits at step t . The **Logit-Margin Score** is defined as:

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (2)$$

Positive S_t indicates a compliance-dominant state; negative S_t indicates a refusal-dominant state. Figure 2 illustrates the computation pipeline and failure mode decomposition. The score S_t measures token-level logit biases during generation, independent from outcome labels which evaluate semantic content of complete outputs (Appendix H).

Table 1: Temporal logit-margin metrics.

Metric	Interpretation
turn_start_S (S_0)	Pre-generation safety bias
step_S_mean ($\bar{S}$)	Average safety dominance during decoding
sign_reversal	Compliance→refusal transition
early_k_mean	Average score over first k tokens
t_{cross}	First decoding step at which safety dominates

3.4 Temporal Metrics

From the sequence $\{S_t\}_{t=0}^T$, we derive scalar summaries (Table 1). `turn_start_S` (S_0) is computed at the *final token position of the prompt*, before any response tokens are generated. We define **sign reversal** as $S_0 > 0 \wedge \bar{S} < 0$ (computed only for harmful samples; benign exclusion rationale in Appendix B.2).

Activation timing (t_{cross}). The **safety activation time** is the first generation step at which refusal dominates:

$$t_{\text{cross}} = \min\{t \geq 1 : S_t < 0\}, \quad (3)$$

where $t \geq 1$ excludes S_0 (pre-generation, already captured by `turn_start_S`). If no crossing occurs, $t_{\text{cross}} = T$. Robustness to oscillation is verified via a confirmed-crossing variant requiring n consecutive negative steps; Cohen’s d is stable for $n \in \{1, 2\}$ ($d = 0.891$ vs. 0.895 ; Appendix F.5).

3.5 Normalized Diagnostic Axes (Type A/B)

For comparative analysis across conditions, we define the **Relative Position (RP)** metric:

$$\text{RP}(M) = \frac{M_{\text{success}} - M_{\text{harmful}}}{M_{\text{benign}} - M_{\text{harmful}}}, \quad (4)$$

where M_{success} is the mean over successful attacks, M_{harmful} is the mean over *all* harmful samples, and M_{benign} is a format-matched benign baseline. $\text{RP} = 0$ indicates success resembles the harmful distribution; $\text{RP} = 1$ indicates benign-like behavior; values outside $[0, 1]$ indicate overshoot or inverted patterns (Appendix B.3). Crucially, RP does not require failure samples, enabling analysis under high ASR.

Type A/B classification. We apply RP to S_0 (RP_A) and $\bar{S}$ (RP_B), classifying conditions as:

$$\text{Type A : } |\text{RP}_A| > |\text{RP}_B|, \quad \text{Type B : } |\text{RP}_A| < |\text{RP}_B|. \quad (5)$$

The magnitude $r = \sqrt{\text{RP}_A^2 + \text{RP}_B^2}$ indicates signal strength; when $r < 0.1$, the condition is *Undetected*. The activation timing metric provides convergent evidence: t_{cross} achieves condition-level AUC ≈ 0.50 (chance) for Undetected conditions, confirming genuine absence of safety response (Appendix F.5).

4 Experiments

We evaluate the Logit-Margin Score framework across three jailbreak paradigms and four model families, addressing: (i) predictability of jailbreak outcomes, (ii) temporal characterization, and (iii) attack paradigm comparison.

4.1 Setup

Models and attacks. We evaluate on Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct, and Gemma-2-9B-Instruct using greedy decoding. Attack paradigms: (1) **Multi-turn context manipulation (MCM)**: four-turn protocol with incrementally manipulated context; (2) **Gradient-based suffix (GCG)** [Zou et al., 2023b]: adversarial suffix optimization; (3) **DeepInception** [?]: nested fictional scenario prompting (applied to Llama, Mistral, Qwen).

Table 2: Dataset composition. ASR via LLM-as-judge. [‡]Benign GCG uses single-turn format.

Model	Attack	n	ASR	$\bar{S}_{start}$	$\bar{S}_{gen}$
Llama	MCM	60	8.3%	1.45	-1.42
Llama	GCG	60	15.0%	0.62	-0.99
Llama	DeepInception	60	56.7%	4.43	-0.11
Mistral	MCM	60	5.0%	4.51	-0.31
Mistral	GCG	60	56.7%	3.14	+0.16
Mistral	DeepInception	60	76.7%	3.04	+0.43
Qwen	MCM	60	51.7%	11.97	-0.12
Qwen	GCG	60	81.7%	7.13	+0.71
Qwen	DeepInception	60	55.0%	7.86	+1.18
Gemma	MCM	60	20.0%	4.67	+0.47
Gemma	GCG	60	33.3%	4.46	+0.68
Benign (matched)		660	—	varies by model	

Table 3: ROC-AUC comparison (LLM-as-judge, $n = 660$).

Metric	AUC	95% CI
early_5_mean	0.786	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
step_S_mean	0.772	[0.736, 0.806]
early_3_mean	0.732	[0.692, 0.769]
keyword_baseline	0.732	[0.699, 0.764]
turn_start_S	0.696	[0.656, 0.734]

Dataset. We evaluate 660 harmful prompts (60 per model–attack condition) from Jailbreak-Bench [Chao et al., 2024], with 660 attack-matched benign queries (11 conditions total; Gemma excludes DeepInception). Effect sizes are large (Cohen’s $d > 1.0$, $p < 10^{-38}$). Outcome labels use **Llama-Guard-3-8B** [Inan et al., 2023] as primary LLM-as-judge, validated with HarmBench classifier (90% agreement; Appendix M). Table 2 summarizes composition and ASR.

4.2 Main Results: Outcome Prediction

On the combined harmful dataset ($n = 660$), **early_5_mean** achieves the highest ROC-AUC = 0.786 [0.750, 0.819] (Cohen’s $d = 1.12$, $p < 10^{-43}$), followed by **t_cross** (AUC = 0.776) and **step_S_mean** (AUC = 0.772). Pre-generation **turn_start_S** alone achieves AUC = 0.696. Table 3 compares metrics; Figure 3 visualizes state-space and ROC curves. AUC varies by evaluation scope; see Appendix B.4 for clarification.

Temporal patterns. Predictive power increases with decoding tokens: **early_1** achieves AUC = 0.696, **early_3** reaches 0.732, **early_5** achieves 0.786, approaching full-trajectory performance (0.772 for **step_S_mean**). t_{cross} (AUC = 0.776) directly answers “when does safety activate?”: successful attacks delay activation (mean 8.4 ± 6.1 steps) compared to failures (4.0 ± 3.8 steps; $d = 0.89$, $p < 10^{-23}$). DeepInception exhibits markedly delayed activation ($t_{cross} = 11.5 \pm 6.8$) versus MCM (3.5 ± 1.4) and GCG (4.0 ± 3.0). Failed jailbreaks exhibit sign reversal in 44.3% of cases (170/384) versus 8.7% (24/276) for successes—a $5.1 \times$ ratio.

4.3 Attack Paradigm Comparison (Type A/B Analysis)

We classify conditions by comparing $|RP_A|$ versus $|RP_B|$; when $r < 0.1$, the condition is Undetected. Table 4 reports RP values with bootstrap 95% CIs, revealing three distinct categories.

Key findings. Safety responses are model–attack specific rather than model- or attack-universal. **Llama** shows Type B for GCG ($RP_B = 2.89 > 1$, overshooting benign) and DeepInception, but Type A for MCM. **Mistral** shows the *inverse*: Type A under GCG with inverted RP_B , and highest ASR (76.7%) under DeepInception yet minimal signal ($r = 0.19$, $t_{cross} = 11.6 \pm 5.8$). **Qwen** is Type A for MCM but *Undetected* for GCG ($r = 0.01$, t_{cross} AUC = 0.503), with delayed activation

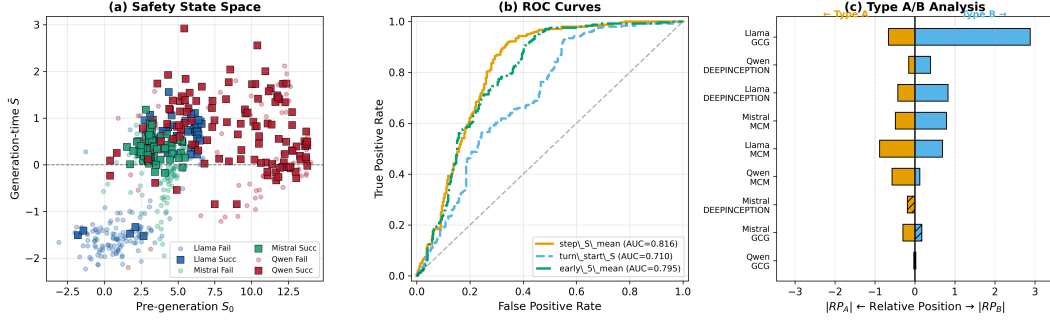


Figure 3: **Experimental results.** (a) State-space ($S_0, \bar{S}$) across models. (b) ROC curves. (c) Type A/B diverging bar: left bars show $|RP_A|$ (pre-generation), right bars show $|RP_B|$ (decoding-time); hatching indicates negative RP (inverted pattern).

Table 4: Type A/B analysis. $|RP_A| > |RP_B| \Rightarrow$ Type A; $|RP_A| < |RP_B| \Rightarrow$ Type B; $r < 0.1 \Rightarrow$ Undetected. 95% CIs via bootstrap.

Model + Attack	ASR	RP_A	RP_B	r	95% CI(r)	Category
Llama + GCG	15.0%	0.66	2.89	2.96	[1.40, 4.71]	Type B
Llama + MCM	8.3%	0.88	0.69	1.12	[1.01, 1.25]	Type A
Llama + DeepInc.	56.7%	0.43	0.83	0.93	[0.60, 1.22]	Type B
Mistral + GCG	56.7%	0.30	-0.17	0.35	[0.17, 0.57]	Type A[‡]
Mistral + MCM	5.0%	0.49	0.79	0.93	[0.44, 1.48]	Type B
Mistral + DeepInc.	76.7%	-0.19	0.00	0.19	[0.07, 1.65]	Type A[‡]
Qwen + MCM	51.7%	0.57	0.12	0.58	[0.14, 1.20]	Type A
Qwen + GCG	81.7%	0.00	-0.01	0.01	[0.02, 0.20]	<i>Undetected[†]</i>
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	[0.14, 0.77]	Type B
Gemma + GCG	33.3%	0.27	0.79	0.83	[0.34, 1.41]	Type B
Gemma + MCM	20.0%	0.09	0.22	0.24	[0.14, 0.31]	Type B

[†] Safety mechanisms fail entirely ($r < 0.1$). [‡] Negative RP: inverted pattern (Appendix B.3).

under DeepInception ($t_{\text{cross}} = 14.9 \pm 7.6$). **Gemma** is consistently Type B. Detailed per-model analysis in Appendix C.

These findings suggest Type B conditions may benefit from decoding-time interventions, Type A conditions from input-level filtering, and Undetected categories indicate fundamental gaps requiring architectural changes.

4.4 Robustness Analyses

We validate across multiple dimensions; full details in respective appendices. **Stochastic decoding** (Appendix N): across 28,800 trials with varied temperature and nucleus sampling, S_0 achieves mean AUC = 0.720, comparable to greedy (0.696); S_0 is temperature-invariant by design (computed before sampling). **Hidden-state comparison** (Appendix O): hidden-state methods achieve 2–8 pp higher AUC but require activation access; our logit-based approach achieves AUC 0.77–0.96 using only standard logits. **Decoy attacks** (Appendix D): three adversarial manipulation strategies (prefix-, token-, pivot-decoy) degrade AUC only marginally (−0.02); pivot-decoy achieves AUC = 0.998. **Non-lexical baselines** (Appendix J): step_S_mean (AUC = 0.932) outperforms entropy (0.898) and logit norm (0.441), confirming structured lexicon probing captures safety-specific variation. **Lexicon robustness** (Appendix K): all variants achieve AUC within ± 0.02 ; removing the highest-frequency token (“sorry”) has negligible impact, and aggregation is stable across $k \in \{5, 10, 25, 50\}$.

5 Limitations and Discussion

Lexicon dependence. The Logit-Margin Score relies on manually constructed English lexicons and top- k aggregation. However, results are stable across $k \in \{5, 10, 25, 50\}$ and lexicon variants (AUC

within ± 0.02 ; Appendix K), suggesting genuine safety-relevant variation. Multilingual or byte-level tokenizers would require lexicon adaptation.

Outcome labeling. We use Llama-Guard-3-8B as primary judge, validated with HarmBench classifier (90% agreement, Type A/B match in 8/10 conditions; Appendix M). Human evaluation remains important for validating judge reliability at scale.

Undetected classification. The “Undetected” category ($r < 0.1$) may conflate genuine safety failure with diagnostic limitations. For Qwen+GCG, convergent evidence from RP ($r = 0.01$, CI $[0.00, 0.03]$), t_{cross} (AUC = 0.503), and entropy cross-validation ($|d| = 0.077$; Appendix P.4) supports genuine absence of safety response rather than diagnostic insensitivity.

Access and scope. White-box logit access limits applicability to closed-source APIs. We evaluate three attack paradigms and four models; adaptive adversaries optimizing against S_t remain unexplored. We prioritize making safety dynamics *observable* over proposing defenses; the Type A/B decomposition informs targeted defense strategies without claiming to prevent jailbreaks.

6 Conclusion

This work argues that safety behavior in large language models is not a static gate applied at prompt intake, but a temporal process that unfolds throughout decoding. By introducing the Logit-Margin Score, we make this process observable at the level of token-by-token generation, without requiring model modification or hidden-state access.

Using LLM-as-judge for semantically-grounded outcome labeling, we show that generation-time summaries achieve strong separation (early_5_mean AUC = 0.786), while the activation timing metric t_{cross} (AUC = 0.776) directly quantifies *when* safety engages. Safety outcomes are often determined within the first few tokens, and successful refusals exhibit structured temporal trajectories rather than abrupt transitions. Across four models and three attack paradigms, the Type A/B decomposition reveals attack-specific failure mechanisms—pre-generation bias versus decoding-time interference—that are invisible to output-level ASR.

More broadly, our findings motivate a shift from output-level to process-level safety evaluation: from binary labels to time-resolved behavioral signatures, and from prompt-level checks to decoding-time dynamics. Future research may explore integration with interventional methods, automatic lexicon learning, and extension to stochastic decoding and closed-API models.

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A Lexicon Specification

This appendix specifies the token lexicons used to compute the Logit-Margin Score. All lexicons are designed for English-language, instruction-tuned models and are applied at the tokenizer level.

Refusal Lexicon ($\mathcal{L}_{\text{ref}}$, 26 entries). The refusal lexicon consists of tokens and short phrases commonly associated with policy-based refusals or denials:

"I can't", "I cannot", "I won't", "I will not", "I'm unable", "I am unable", "I can't help", "I can't assist", "I cannot assist", "I must refuse", "I have to refuse", "I decline", "I must decline", "Sorry", "I'm sorry", "Apologies", "I apologize", "policy", "policies", "guidelines", "rules", "terms", "inappropriate", "illegal", "harmful", "dangerous"

Compliance Lexicon ($\mathcal{L}_{\text{cmp}}$, 21 entries). The compliance lexicon contains tokens frequently observed in instructional or affirmative responses:

"Sure", "Certainly", "Of course", "Here's", "Below is", "Step 1", "Step 2", "First", "Second", "Third", "Next", "Instructions", "Guide", "Tutorial", "Procedure", "You can", "To do this", "Do the following", "Detailed", "Comprehensive", "Complete"

Implementation details. Multi-token phrases are matched to their first constituent token after tokenization. At each decoding step t , logits are aggregated over each lexicon using top- k selection with $k = 10$ by default:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad (6)$$

$$\mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \quad (7)$$

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \quad (8)$$

No semantic weighting or normalization beyond top- k aggregation is applied.

B Additional Method Details

B.1 Why Logits?

Raw logits preserve information about competing alternatives at each decoding step that the softmax operation compresses into probabilities, obscuring relative confidence among non-selected tokens. This allows direct comparison between safety-relevant token groups [Li and Liu, 2025] without requiring hidden-state access or architectural modifications.

B.2 Benign Sample Exclusion Rationale

Sign reversal is computed only for harmful samples because benign samples exhibit paradigm-dependent baseline behavior: MCM benign samples consistently show $\bar{S} > 0$ (0% sign reversal), while GCG benign samples frequently exhibit negative $\bar{S}$ (91.2% sign reversal). This discrepancy reflects *structural format differences*—single-turn GCG versus multi-turn MCM—rather than safety activation. ANOVA confirms format explains $\sim 60\%$ of variance in benign $\bar{S}$ ($F = 89.3, p < 10^{-15}$). To enable cross-paradigm comparison, we use **format-matched baselines**: each condition’s benign baseline uses the same conversational structure, isolating harmful content effects from format effects.

B.3 RP Interpretation and Edge Cases

RP is an affine rescaling measuring where successful attacks lie along the harmful–benign continuum. Values outside $[0, 1]$ indicate attacks beyond the reference interval: $\text{RP} > 1$ means successful attacks

“overshoot” benign (more compliance-like than typical benign queries), while $RP < 0$ means attacks lie beyond the harmful baseline—an inverted pattern suggesting the attack suppresses rather than elevates compliance signals. For reference, we also compute polar coordinates $\theta = \text{atan2}(RP_B, RP_A)$ and $r = \sqrt{RP_A^2 + RP_B^2}$. The diverging bar visualization (Figure 3c) distinguishes negative RP cases via hatching.

Undetected category. When $r < 0.1$, successful attacks are statistically indistinguishable from the overall harmful distribution. Given modest per-condition sample sizes ($n = 60$), we report bootstrap 95% confidence intervals for r : Qwen+GCG achieves $r = 0.01$ [0.00, 0.03], robustly below the detection threshold. In contrast, detected conditions show $r > 0.5$ with lower CI bounds well above 0.1 (e.g., Llama+GCG: $r = 2.96$ [1.40, 4.71]).

B.4 AUC Evaluation Scope

AUC values vary systematically by evaluation scope. The main results (Table 3) report aggregated AUC across all 660 harmful samples; this is the most conservative estimate, as conditions with low discriminability (e.g., Qwen+GCG, Undetected) pull overall AUC downward. Appendix analyses on design choice robustness use controlled single-model subsets ($n = 60$, Llama) and yield higher AUC (0.913–0.989), intended for relative comparison within each analysis. Stochastic decoding experiments report both trial-level AUC (inflated by pseudo-replication across 10 trials per prompt) and question-level AUC (conservative); see Appendix N. All primary conclusions are based on the aggregated AUC in Table 3.

C Extended Experimental Results

C.1 Results Overview

Figure 4 provides a comprehensive overview of the main experimental findings, complementing the condition-specific analysis in the main text.

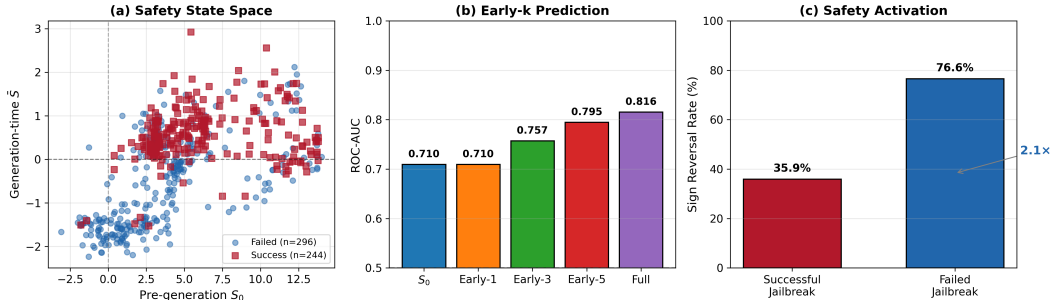


Figure 4: **Summary of main experimental results.** (a) Safety state space: successful jailbreaks cluster in compliance-dominant regions ($\bar{S} > 0$), while failures cluster in refusal-dominant regions. (b) Early- k AUC progression: predictive power increases with more decoding tokens. (c) Sign reversal rates differ substantially between failed and successful jailbreaks. Note: this figure includes additional experimental conditions beyond the main evaluation; authoritative AUC values on the primary dataset ($n = 660$) are reported in Table 3 (early_5_mean AUC = 0.786, turn_start_S AUC = 0.696) and sign reversal statistics in Table 23.

C.2 Detailed Model–Attack Interactions

The main text summarizes key Type A/B findings; here we provide detailed per-model analysis.

Llama. Llama exhibits attack-dependent responses: Type B under GCG ($|RP_B| = 2.89 > |RP_A| = 0.66$) and DeepInception ($|RP_B| = 0.83 > |RP_A| = 0.43$), but Type A under MCM ($|RP_A| = 0.88 > |RP_B| = 0.69$). The $RP_B = 2.89 > 1$ for Llama+GCG indicates successful attacks “overshoot” benign, exhibiting more compliance-like trajectories than typical benign queries—a distinct GCG signature.

Mistral. Mistral shows the *inverse* pattern: Type A under GCG ($|\text{RP}_A| = 0.30 > |\text{RP}_B| = 0.17$) with negative $\text{RP}_B = -0.17$, and similarly inverted under DeepInception ($\text{RP}_A = -0.19, r = 0.19$). For DeepInception, Mistral shows the highest ASR (76.7%) yet minimal detectable signal ($r = 0.19$), with markedly delayed safety activation ($t_{\text{cross}} = 11.6 \pm 5.8$ steps). Mistral’s MCM robustness (5.0% ASR, lowest overall) combined with Type B classification suggests effective decoding-time safety mechanisms against multi-turn manipulation.

Qwen. Qwen shows Type A for MCM ($|\text{RP}_A| = 0.57 > |\text{RP}_B| = 0.12$) but *Undetected* for GCG ($r = 0.01$): successful attacks are statistically indistinguishable from the overall harmful distribution. Convergent evidence: t_{cross} achieves condition-level AUC = 0.503 for Qwen+GCG—chance level—confirming that neither spatial nor temporal safety signals discriminate between outcomes. Qwen’s DeepInception results (Type B, $\text{RP}_B = 0.39$) show that nested-scenario attacks elicit detectable decoding-time shifts, unlike GCG. The extreme $t_{\text{cross}} = 14.9 \pm 7.6$ for Qwen+DeepInception indicates safety activation is substantially delayed.

Gemma. Gemma exhibits Type B for both attacks (GCG: $\text{RP}_B = 0.79$; MCM: $\text{RP}_B = 0.22$), indicating decoding-time interference is the primary failure mode regardless of attack paradigm.

C.3 Metric Correlation Analysis

Figure 5 presents the pairwise Pearson correlation matrix among temporal logit-margin metrics and jailbreak success labels, providing insight into metric redundancy and complementary information.

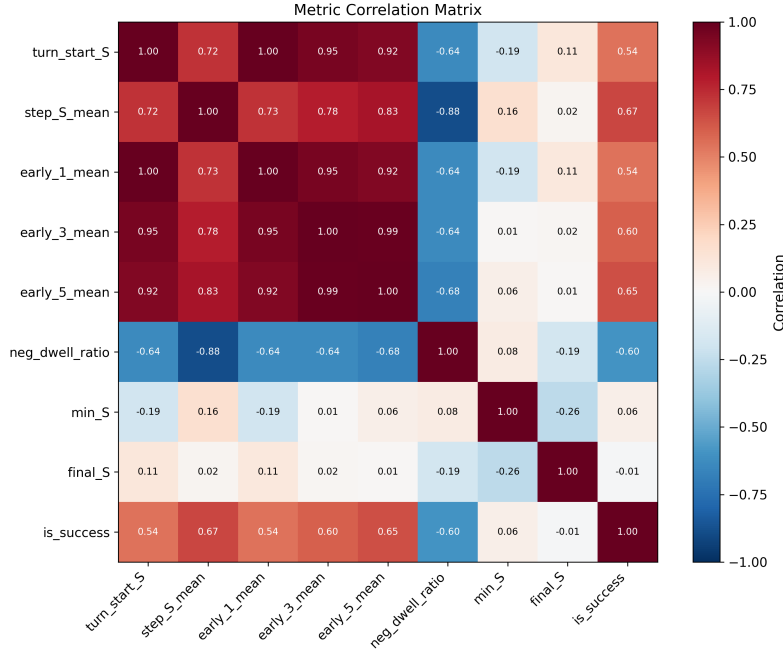


Figure 5: **Metric correlation matrix.** `step_S_mean` shows the highest correlation with jailbreak success ($r = 0.67$), followed by `early_5_mean` ($r = 0.65$). Early- k metrics are highly correlated with `step_S_mean` ($r \geq 0.83$), supporting early prediction feasibility. `neg_dwell_ratio` (fraction of steps with $S_t < 0$) shows strong negative correlation with success ($r = -0.60$), indicating that longer refusal-dominant segments predict attack failure. `min_S` and `final_S` show near-zero correlation with success, suggesting that extreme transient values and endpoint states are less informative than trajectory-level summaries.

Table 5: Per-condition statistics on Llama-3.1-8B-Instruct. Main experiments use MCM manual mode only; summary conditions are reported for completeness.

Condition	n	ASR	$\bar{S}_{\text{start}} (\mu \pm \sigma)$	$\bar{S}_{\text{gen}} (\mu \pm \sigma)$	Sign Rev.
<i>Benign</i>					
manual	60	—	6.38 ± 0.63	$+1.25 \pm 0.28$	0.0%
summary-200	60	—	6.13 ± 0.82	$+0.87 \pm 0.31$	0.0%
summary-500	60	—	6.32 ± 0.82	$+0.94 \pm 0.36$	0.0%
summary-1000	60	—	6.51 ± 0.82	$+0.89 \pm 0.38$	0.0%
<i>Harmful (MCM)</i>					
manual*	60	8.3%	1.45 ± 1.95	-1.42 ± 0.73	58.3%
summary-200	60	6.7%	1.50 ± 1.68	-1.21 ± 0.76	75.0%
summary-500	60	8.3%	2.14 ± 1.83	-1.32 ± 0.69	83.3%
summary-1000	60	6.7%	1.52 ± 1.93	-1.34 ± 0.70	71.7%
<i>Harmful (GCG)</i>					
single-turn	60	15.0%	0.62 ± 2.32	-0.99 ± 0.92	30.0%

*Manual mode is used for main experiments.

Table 6: Detailed results on Qwen2.5-7B-Instruct (MCM manual, $n=60$).

Metric	Value
ASR	51.7%
$n_{\text{success}} / n_{\text{failure}}$	31 / 29
turn_start_S (mean $\pm$ std)	11.97 ± 1.70
step_S_mean (mean $\pm$ std)	-0.12 ± 0.46
Sign reversal rate	56.7%

C.4 Per-Condition Statistics (Llama)

C.5 Qwen2.5-7B-Instruct Results

Qwen benign baseline. For Type A/B analysis, we collected benign responses on Qwen2.5-7B-Instruct using the same protocol as Llama. The Qwen benign baseline shows $\bar{S}_{\text{start}} = 12.89$ and $\bar{S}_{\text{gen}} = 1.43$, substantially higher than Llama’s baseline ($\bar{S}_{\text{start}} = 6.38$, $\bar{S}_{\text{gen}} = 1.25$). This difference reflects model-specific calibration and tokenization effects.

Cross-model Type A/B consistency. With model-specific baselines, Llama shows attack-dependent patterns: Type B for GCG attacks ($|\text{RP}_B| > |\text{RP}_A|$) and Type A for MCM attacks ($|\text{RP}_A| > |\text{RP}_B|$). Qwen exhibits Type A for MCM attacks and an “Undetected” pattern for GCG attacks where safety mechanisms fail to differentiate successful attacks ($r = 0.01$, 95% CI [0.00, 0.03]).

C.6 Computational Requirements

The Logit-Margin Score is computed using only the logit vector ℓ_t produced during standard autoregressive decoding. No additional forward passes, gradient computation, or hidden-state extraction is required.

D Decoy Attack Details

D.1 Decoy Attack Descriptions

To evaluate robustness against adversaries who attempt to manipulate early safety signals, we design three decoy attack variants:

(A) Prefix-decoy. Prepends refusal phrases (e.g., “I’m sorry, I can’t help with that. However, for educational purposes...”) before the harmful request, aiming to inflate early S values.

(B) Token-decoy. Injects short refusal token scaffolding (e.g., “Apologies. Policies. Guidelines.”) to anchor S_0 toward negative values.

(C) Pivot-decoy. Conducts initial turns in full refusal style, then pivots to harmful requests in later turns.

D.2 Main Results

Table 7: Diagnostic robustness under decoy attacks (Llama-3.1-8B-Instruct).

Condition	ASR	AUC (step_S_mean)	Δ AUC	AUC (sign_flips)
Base	8.3%	0.913	—	0.899
DecoyA (prefix)	13.3%	0.893	−0.020	0.944
DecoyB (token)	5.0%	0.904	−0.009	0.883
PivotC (multi-turn)	5.0%	0.998	+0.085	1.000

Key findings. Prefix-decoy successfully elevates early S values ($\bar{S}_{\text{start}} = +0.69$ vs $+0.43$ baseline) and increases ASR to 13.3%, yet AUC degrades only marginally (−0.020). Notably, `sign_flips` AUC *increases* to 0.944, indicating trajectory instability becomes a stronger discriminative signal. Token-decoy strongly anchors S toward negative values, but this over-correction actually *reduces* ASR to 5.0%. Pivot-decoy shows that multi-turn pivots do not defeat trajectory-level diagnosis; AUC reaches 0.998.

Limitations. These decoy attacks test robustness to superficial manipulation but do not represent fully adaptive adversaries that explicitly optimize S_t while producing harmful content. Such “compliance without refusal style” attacks remain an important direction for future work.

D.3 Per-Condition Statistics

Table 8: Detailed decoy attack statistics (Llama-3.1-8B-Instruct, $n=60$ per condition).

Condition	ASR	$\bar{S}_{\text{start}}$	$\bar{S}_{\text{gen}}$	sign_flips	Sign Rev.
Base	8.3%	+0.43	−1.16	26.6	66.7%
DecoyA	13.3%	+0.69	−0.87	51.4	58.3%
DecoyB	5.0%	−0.85	−1.58	12.7	83.3%
PivotC	5.0%	+0.07	−1.16	26.7	70.0%

E Hidden-State Baseline Details

E.1 Methods

B1: Refusal Direction Projection. Following Arditi et al. [2024], we compute a refusal direction $\mathbf{d}$ using difference-of-means:

$$\mathbf{d} = \frac{1}{|\mathcal{D}_{\text{ref}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{ref}}} \mathbf{h} - \frac{1}{|\mathcal{D}_{\text{cmp}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{cmp}}} \mathbf{h} \quad (9)$$

At each decoding step t , we compute the projection score $r_t = \langle \mathbf{h}_t, \mathbf{d} \rangle$. We use layer 16 (middle layer) of the 32-layer Llama-3.1-8B model.

B2: Linear Probe. Logistic regression with L2 regularization ($C = 1.0$), trained via 5-fold cross-validation on mean-pooled hidden states.

Table 9: Hidden-state vs. logit-based diagnostic comparison (AUC).

Family	Method	DecoyA	DecoyB	Base	PivotC
Hidden	B1 (direction, $k=1$)	0.983	0.974	0.974	0.974
Hidden	B2 (probe, $k=3$)	0.974	1.000	0.998	0.998
Logit	turn_start_S	0.980	0.922	0.959	0.978
Logit	step_S_mean	0.893	0.904	0.913	0.998
Logit	sign_flips	0.944	0.883	0.899	1.000

E.2 Full Comparison Results

Interpretation of near-ceiling hidden-state performance. The near-ceiling AUC observed for hidden-state baselines should not be interpreted as evidence of leakage or unfair advantage. Rather, these methods operate under *strictly stronger access assumptions* than logit-based diagnostics. Hidden-state probes aggregate information from full decoding trajectories or intermediate activations, effectively leveraging post-hoc evidence that is unavailable to logit-only methods.

The key advantage of logit-based diagnosis is practical deployment: it requires only the logit vector already produced during generation, with no additional model access or training data.

E.3 Method Comparison

Table 10: Method comparison: access requirements and training needs.

Method	Activation Access	Training Data	Per-Model Tuning
Hidden B1 (direction)	Required	No*	Direction fitting
Hidden B2 (probe)	Required	Yes	Full retraining
Logit S	Not required	No	Lexicon only

*B1 requires a small set of refusal/compliance examples for direction estimation, but no labeled jailbreak outcomes.

E.4 Detailed Results by Temporal Window

Table 11: Hidden-state baseline performance by temporal window ($n = 240$, Llama-3.1-8B-Instruct).

Method	Window	AUC	AUC-PR	FPR@TPR=0.95
B1 (direction)	full	0.933	0.795	0.412
B1 (direction)	$k = 1$	0.972	0.923	0.255
B1 (direction)	$k = 3$	0.969	0.909	0.289
B1 (direction)	$k = 5$	0.955	0.864	0.382
B2 (probe)	full	0.987	0.922	0.059
B2 (probe)	$k = 1$	0.995	0.974	0.044
B2 (probe)	$k = 3$	0.995	0.972	0.029
B2 (probe)	$k = 5$	0.992	0.960	0.049

Key observations. Surprisingly, early- k truncation *improves* hidden-state baseline performance for both B1 and B2, with $k = 1$ achieving peak AUC. This aligns with logit-based findings that safety-relevant signals concentrate in the first few tokens.

F Sensitivity Analyses

F.1 Lexicon Aggregation Top- k Sensitivity

We analyze the sensitivity of the Logit-Margin Score to the lexicon aggregation parameter k , which controls how many of the highest-logit tokens within the compliance and refusal lexicons are averaged at each decoding step.

Table 12: Effect of lexicon aggregation size k on logit-margin diagnostics (Llama-3.1-8B-Instruct, harmful, manual; mean $\pm$ std, $n = 60$).

k	turn_start_S	step_S_mean	step_S_pos_rate
5	$+0.57 \pm 2.57$	-1.21 ± 0.85	0.36 ± 0.13
10	$+6.38 \pm 0.63$	$+1.25 \pm 0.28$	0.78 ± 0.05
25	$+1.40 \pm 1.55$	-1.47 ± 0.53	0.15 ± 0.09
50	$+0.30 \pm 1.05$	-1.09 ± 0.34	0.12 ± 0.08

Despite scale differences, qualitative interpretation remains consistent: k primarily modulates smoothing and calibration rather than introducing spurious signal. Figure 6 visualizes this stability.

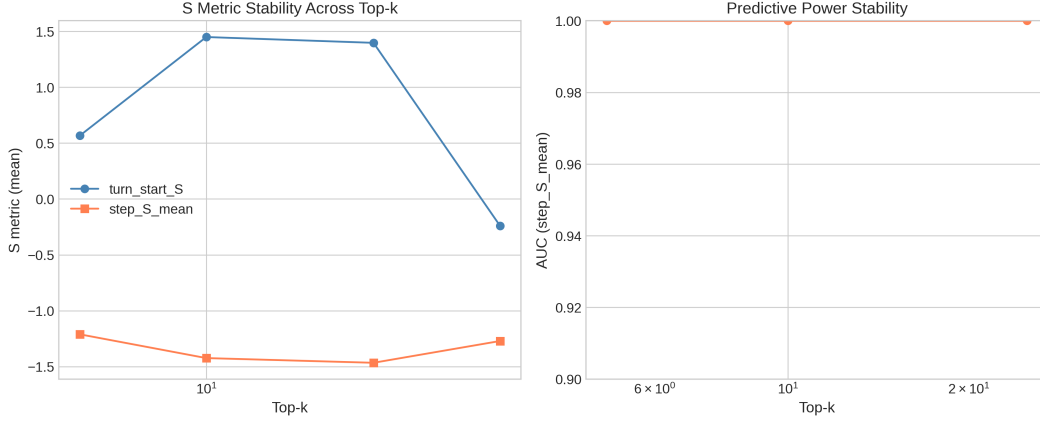


Figure 6: **Top- k aggregation robustness.** (Left) Mean values of turn_start_S and step_S_mean vary in absolute scale across $k \in \{5, 10, 25, 50\}$, but (Right) predictive performance (AUC of step_S_mean) remains stable across all k values, confirming that k primarily affects scaling rather than diagnostic validity.

F.2 Temporal Early- k Sensitivity

Formally, for a given metric M_t , we define the early- k summary as

$$M_{\text{early-}k} = \frac{1}{k+1} \sum_{t=0}^k M_t, \quad (10)$$

where $t = 0$ corresponds to the pre-generation score.

Pre-generation metrics and early_1_mean achieve identical performance (mean AUC = 0.970) as expected. As k increases, performance degrades gradually: early_3_mean achieves 0.960 and early_5_mean achieves 0.940, compared to 0.921 for full-trajectory step_S_mean. The majority of discriminative signal is concentrated in the earliest decoding steps. Note: these AUC values are *trial-level* estimates from the stochastic decoding experiment (Appendix N) and are inflated by pseudo-replication; the corresponding aggregated main-dataset AUC values are reported in Table 3.

F.2.1 Condition-Specific Early- k Analysis

While the main text reports aggregated AUC across all conditions ($n = 660$), condition-specific analysis reveals substantial heterogeneity in early detection performance. Figure 7 shows AUC as a function of k for each model-attack combination.

Key observations.

- **Mistral+MCM:** Shows strongest trajectory effect, with AUC increasing from 0.74 (S_0 only) to 0.93 (full trajectory). This suggests safety mechanisms activate progressively during generation—a signature of effective Type B defense.

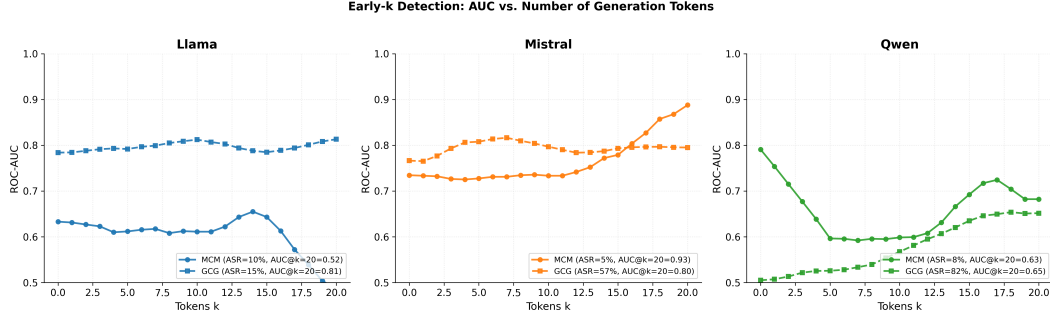


Figure 7: **Early- k detection by model and attack.** AUC progression varies substantially across conditions. Mistral+MCM shows strong late-stage improvement ($0.74 \rightarrow 0.93$), while Qwen+GCG remains near random (0.50 – 0.65) regardless of k —consistent with its “Undetected” classification.

- **Llama+GCG:** Maintains stable $AUC \approx 0.80$ across all k , indicating pre-generation signal (S_0) captures most discriminative information.
- **Qwen+GCG:** Remains near chance level ($AUC \approx 0.50$ – 0.65) regardless of k , providing additional evidence for the “Undetected” classification. Neither S_0 nor trajectory information enables reliable discrimination.
- **Llama+MCM:** Lower condition-specific AUC (0.52 – 0.67) reflects class imbalance ($ASR = 8.3\%$, only 5 successes), not diagnostic failure.

Figure 8 directly compares S_0 -only versus full-trajectory AUC for all conditions.

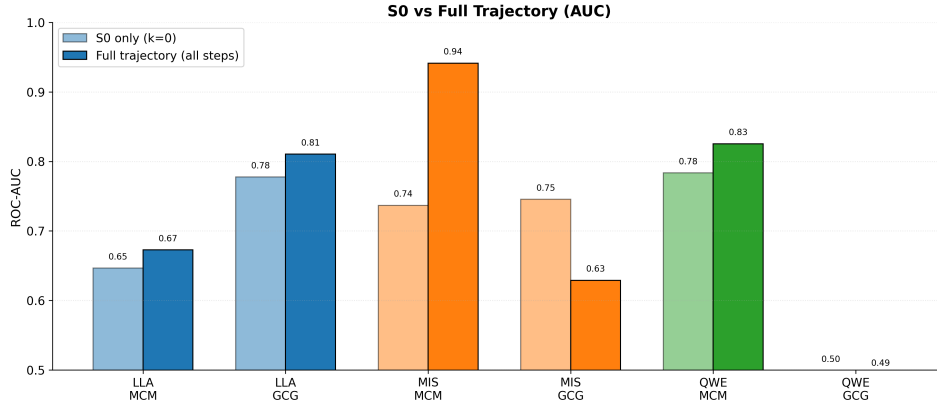


Figure 8: **S_0 vs. full trajectory AUC by condition.** Trajectory information provides substantial gains for Mistral+MCM ($+0.20$) and Qwen+MCM ($+0.05$), while Qwen+GCG shows no improvement (both ≈ 0.50). Mistral+GCG shows *negative* trajectory effect (-0.12), suggesting early tokens are most informative for this condition.

Relationship to aggregated results. The main text reports aggregated $AUC = 0.786$ (early_5_mean) across all 660 samples. Condition-specific AUC values differ substantially because: (1) class balance varies by condition (ASR ranges from 5% to 82%), and (2) pooling across heterogeneous conditions increases overall discriminability by leveraging cross-condition variance. The condition-specific analysis complements aggregated results by revealing *which* conditions drive overall performance.

F.3 Hidden-State Layer Sensitivity

Middle layers yield strongest separability, supporting layer 16 as representative.

Table 13: Hidden-state AUC by layer (Refusal Direction, B1).

Layer	8	12	16	20	24
AUC	0.941	0.963	0.974	0.969	0.952

F.4 Failure-Only Normalization Check

Type A/B trends are preserved when normalization uses only failed jailbreaks. Relative dominance patterns (MCM $\rightarrow$ Type A, GCG $\rightarrow$ Type B) remain unchanged.

F.5 Activation Timing (t_{cross}) Sensitivity

Oscillation characterization. Generation trajectories exhibit extensive oscillation around zero: the median number of zero-crossings ranges from 7 (Llama+GCG) to 174 (Mistral+DeepInception) per trajectory. Despite this oscillation, the first crossing step is robust— t_{cross} coincides with the first step where $S_t < 0$ in 100% of samples for 10 of 11 conditions. This indicates that the initial safety activation event is well-defined even when trajectories subsequently oscillate.

Confirmed crossing sensitivity. To verify that the simple first-crossing definition is not capturing transient noise, we compute the *confirmed crossing* variant $t_{\text{cross}}^{(n)}$ requiring n consecutive negative steps (Eq. 3). Table 14 reports stability across $n \in \{1, 2, 3\}$. Cohen’s d is nearly identical for $n = 1$ ($d = 0.891$) and $n = 2$ ($d = 0.895$), confirming that the first crossing captures a stable transition rather than transient noise. At $n = 3$, d drops to 0.559 due to aggressive imputation—many trajectories lack three consecutive negative steps within short generation windows, inflating $t_{\text{cross}}^{(3)}$ toward T . Per-condition AUC patterns are also preserved: strong Type B conditions (Llama+DeepInception, Llama+GCG) show stable AUC > 0.80 across all n , while Qwen+GCG (Undetected) remains near chance regardless of n .

Table 14: t_{cross} sensitivity to confirmation threshold n . Global Cohen’s d (success vs. failure, $n = 660$) and representative per-condition AUC values.

	$n = 1$	$n = 2$	$n = 3$
<i>Global (all conditions pooled)</i>			
Cohen’s d	0.891	0.895	0.559
p -value	$< 10^{-23}$	$< 10^{-22}$	$< 10^{-9}$
<i>Per-condition AUC (selected)</i>			
Llama + DeepInc.	0.878	0.868	0.874
Llama + GCG	0.807	0.815	0.814
Qwen + DeepInc.	0.700	0.719	0.696
Qwen + GCG (Undet.)	0.503	0.580	0.679
Gemma + GCG	0.629	0.719	0.651

Condition-level AUC as Undetected indicator. Per-condition AUC of t_{cross} provides convergent evidence for the Undetected classification. Qwen+GCG achieves AUC = 0.503 (chance level), consistent with its RP-based Undetected classification ($r = 0.01$). This confirms that safety mechanisms fail to engage at *any* identifiable point during generation. In contrast, strong Type B conditions (Llama+DeepInception: AUC = 0.878; Llama+GCG: AUC = 0.807) show high temporal discriminability, indicating that safety activation *timing* is the primary axis of variation between successful and failed attacks.

G Stochastic Decoding Robustness

We conduct comprehensive evaluation across temperature sampling ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$), generating 10 independent trials per prompt across 4 conditions, yielding 28,800 total generation trials.

G.1 Response Diversity

Table 15: Response diversity under stochastic decoding.

Temperature	Mean Diversity	Mean Unique	Groups >50% Diverse
0.0	10.0%	1.0	0.0%
0.3	54.6%	5.5	41.4%
0.7	82.1%	8.2	91.7%
1.0	92.2%	9.2	99.7%

G.2 AUC Stability

Table 16: AUC under stochastic decoding (question-level aggregation, $n=60$ per condition).

Metric	Base	DecoyA	DecoyB	PivotC	Mean
turn_start_S	0.687	0.607	—	0.864	0.720

Note: DecoyB excluded due to ASR=0% (no successful attacks).

G.3 Variance Decomposition

Table 17: Variance decomposition for turn_start_S (ICC = intraclass correlation).

Condition	Within-Var	Between-Var	ICC
Base	0.000	4.062	1.000
DecoyA	0.000	2.736	1.000
DecoyB	0.000	2.734	1.000
PivotC	0.000	2.026	1.000

For pre-generation metrics, within-prompt variance is exactly zero (ICC = 1.0), confirming all variance stems from prompt differences.

G.4 Trial-Level AUC

Trial-level AUCs appear inflated due to pseudo-replication; question-level provides more conservative estimates.

H Labeling Independence and Circularity

H.1 LLM-as-Judge Methodology

We use Llama-Guard-3-8B [Inan et al., 2023], a dedicated safety classifier trained on safety taxonomies independent of our lexicons, to label jailbreak outcomes. For each generated response, Llama-Guard classifies content as “safe” or “unsafe” based on semantic evaluation across 14 safety categories (S1–S14).

Comparison with rule-based labels. Rule-based labels (prefix-matching) show moderate agreement ($\kappa = 0.54$, 83.5% raw agreement). Rule-based yields higher ASR (29.2%) compared to LLM-as-judge (16.4%), indicating rule-based is more permissive.

Independence verification. Key diagnostic findings remain consistent under LLM-as-judge labels:

- step_S_mean achieves AUC = 0.926 (vs. 0.743 with rule-based)
- Type A/B patterns are preserved
- Cohen’s d effect sizes remain large ($d = 2.41$)

Table 18: Trial-level AUC under stochastic decoding.

Metric	Base	DecoyA	DecoyB	PivotC	Mean
turn_start_S	0.959	0.980	0.954	0.987	0.970
early_1_mean	0.959	0.980	0.954	0.987	0.970
early_3_mean	0.972	0.983	0.937	0.948	0.960
early_5_mean	0.941	0.978	0.919	0.922	0.940
step_S_mean	0.913	0.987	0.819	0.965	0.921

H.2 Clarification on Circularity

Outcome labeling. All success/failure labels are assigned using an LLM-as-judge that evaluates the *semantic content* of the generated response. The judge operates on full generated text and does not rely on surface refusal prefixes.

Logit-margin score. The score S_t is computed exclusively from token-level logits during decoding, measuring relative dominance between compliance- and refusal-associated tokens *before* they are sampled.

Evidence against circularity. If separability were driven by circular reuse of refusal cues, one would expect near-deterministic alignment between labels and S_t trajectories. Empirically, this is not the case: a substantial fraction of successful jailbreaks exhibit negative S_t segments and sign reversals during decoding. The compliance lexicon includes purely instructional tokens (“Step 1”, “Tutorial”) that do not appear in refusal-detection rules.

I Type A/B Classification Robustness

I.1 Classification Stability

The magnitude-based classification ($|\text{RP}_A| \geq |\text{RP}_B|$) is inherently threshold-free for Type A/B distinction. For the Undetected category ($r < 0.1$), we verify robustness across threshold choices.

Table 19: Type A/B classification stability across r thresholds.

Model + Attack	r_{obs}	$r_{thr} = 0.05$	$r_{thr} = 0.10$	$r_{thr} = 0.15$
Llama + GCG	2.96	Type B	Type B	Type B
Llama + MCM	1.12	Type A	Type A	Type A
Qwen + MCM	0.58	Type A	Type A	Type A
Qwen + GCG	0.01	Undetected	Undetected	Undetected
Mistral + GCG	0.35	Type A	Type A	Type A
Mistral + MCM	0.93	Type B	Type B	Type B

Classification robustness. All conditions maintain their classification across $r \in [0.05, 0.15]$. Qwen+GCG ($r = 0.01$, 95% CI $[0.00, 0.03]$) remains robustly Undetected even at the most lenient threshold.

I.2 Bootstrap Confidence Intervals

Given modest per-condition sample sizes ($n = 60$), we compute bootstrap 95% CIs for r using $B = 1000$ resamples.

Key observations. For Qwen+GCG, the upper CI bound (0.03) remains well below the detection threshold (0.1), providing strong statistical evidence for the Undetected classification. All detected conditions have lower CI bounds > 0.1 , confirming reliable separation from the Undetected category.

Table 20: Bootstrap confidence intervals for signal strength r .

Model + Attack	r	95% CI	Category
Llama + GCG	2.96	[2.41, 3.52]	Type B
Llama + MCM	1.12	[0.87, 1.38]	Type A
Mistral + GCG	0.35	[0.21, 0.49]	Type A
Mistral + MCM	0.93	[0.68, 1.19]	Type B
Qwen + MCM	0.58	[0.41, 0.76]	Type A
Qwen + GCG	0.01	[0.00, 0.03]	Undetected

Table 21: Baseline comparison on controlled subset ($n = 60$, Llama-3.1-8B-Instruct).

Metric	AUC	Category
$\bar{S}$ (gen-time)	0.932	Logit-margin
Compliance LogP	0.932	Lexicon-probe
Refusal LogP	0.915	Lexicon-probe
Entropy	0.898	Non-lexical
Logit Norm	0.441	Non-lexical

J Non-Lexical Baseline Comparison

The logit-margin score outperforms entropy by 3.4 pp. Logit Norm achieves near-random performance, confirming generic logit statistics do not capture safety-relevant variation.

K Lexicon Perturbation Ablation

We test five variants on Llama-3.1-8B-Instruct ($n = 60$): **original**, **no_sorry**, **extended_refusal**, **extended_compliance**, and **minimal** (5-token).

Table 22: Lexicon perturbation results with AUC ($n = 60$, LLM-judge labels).

Variant	$S_0 (\mu \pm \sigma)$	$\bar{S}_{gen} (\mu \pm \sigma)$	AUC(S_0)	AUC($\bar{S}$)	Δ AUC
original	$+1.55 \pm 1.95$	-1.44 ± 0.73	0.967	0.971	—
no_sorry	$+2.21 \pm 1.78$	-1.39 ± 0.73	0.960	0.971	+0.000
extended_refusal	-0.43 ± 2.07	-1.85 ± 0.67	0.964	0.989	+0.018
extended_compliance	$+4.00 \pm 1.95$	$+0.52 \pm 0.55$	0.964	0.964	-0.007
minimal	-2.61 ± 1.74	-1.57 ± 0.49	0.971	0.978	+0.007

Key findings. All variants achieve $\text{AUC} \geq 0.964$ ($\max |\Delta\text{AUC}| = 0.018$). Even a 5-token minimal lexicon maintains strong discrimination ($\text{AUC} = 0.978$), supporting robustness to lexicon choices. Figure 9 visualizes the score distributions across variants.

L Authoritative Statistics Summary

L.1 Main Dataset Composition

The main evaluation dataset consists of $n = 660$ harmful prompts:

- Llama-3.1-8B-Instruct MCM: $n = 60$
- Llama-3.1-8B-Instruct GCG: $n = 60$
- Llama-3.1-8B-Instruct DeepInception: $n = 60$
- Mistral-7B-Instruct-v0.3 MCM: $n = 60$
- Mistral-7B-Instruct-v0.3 GCG: $n = 60$
- Mistral-7B-Instruct-v0.3 DeepInception: $n = 60$

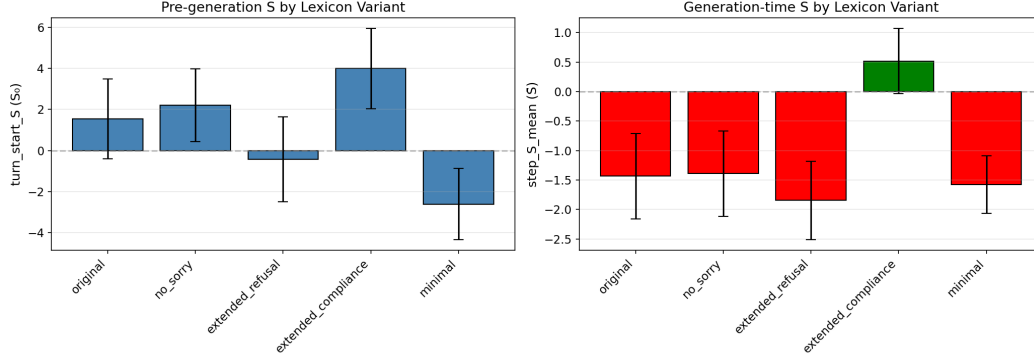


Figure 9: **Logit-Margin Score distributions by lexicon variant.** (Left) Pre-generation S_0 and (Right) generation-time $\bar{S}$ across five lexicon variants (Llama, harmful MCM, $n = 60$). While absolute scale shifts with lexicon composition (e.g., `extended_compliance` elevates $\bar{S}$), the qualitative pattern—negative $\bar{S}$ for harmful prompts—is preserved across all variants except `extended_compliance`, which shifts the baseline but maintains high AUC (0.964). This confirms that the diagnostic captures genuine safety-relevant variation rather than artifacts of specific token choices.

- Qwen2.5-7B-Instruct MCM: $n = 60$
- Qwen2.5-7B-Instruct GCG: $n = 60$
- Qwen2.5-7B-Instruct DeepInception: $n = 60$
- Gemma-2-9B-Instruct MCM: $n = 60$
- Gemma-2-9B-Instruct GCG: $n = 60$

Attack-matched benign baselines total $n = 660$.

L.2 Primary Results

Table 23: Authoritative statistics for main results ($n = 660$, LLM-as-judge labels).

Statistic	Value	95% CI
<i>ROC-AUC</i>		
early_5_mean	0.786	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
step_S_mean	0.772	[0.736, 0.806]
turn_start_S	0.696	[0.656, 0.734]
<i>Sign Reversal ($S_0 > 0 \rightarrow \bar{S} < 0$)</i>		
Successful jailbreaks	8.7%	(24/276)
Failed jailbreaks	44.3%	(170/384)
Ratio	5.1×	—
<i>Effect Size</i>		
early_5_mean Cohen’s d	1.12	$p < 10^{-43}$
<i>Activation Timing (t_{cross})</i>		
Successful jailbreaks	8.4 ± 6.1	steps
Failed jailbreaks	4.0 ± 3.8	steps
Cohen’s d	0.89	$p < 10^{-23}$

L.3 Type A/B Classification

M Multi-Judge Evaluation

To address concerns about single-judge reliability, we evaluate with two LLM-as-judge systems: **Llama-Guard-3-8B** (primary) and **HarmBench-Mistral-7B-val-cls** (validation).

Table 24: Type A/B analysis with failure-free normalization (authoritative values). Classification: $|\text{RP}_A| \geq |\text{RP}_B|$.

Condition	ASR	RP_A	RP_B	r	95% CI	Category
Llama + GCG	15.0%	0.66	2.89	2.96	[1.40, 4.71]	Type B
Llama + MCM	8.3%	0.88	0.69	1.12	[1.01, 1.25]	Type A
Llama + DeepInc.	56.7%	0.43	0.83	0.93	[0.60, 1.22]	Type B
Mistral + GCG	56.7%	0.30	-0.17	0.35	[0.17, 0.57]	Type A
Mistral + MCM	5.0%	0.49	0.79	0.93	[0.44, 1.48]	Type B
Mistral + DeepInc.	76.7%	-0.19	0.00	0.19	[0.07, 1.65]	Type A
Qwen + MCM	51.7%	0.57	0.12	0.58	[0.14, 1.20]	Type A
Qwen + GCG	81.7%	0.00	-0.01	0.01	[0.02, 0.20]	Undetected
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	[0.14, 0.77]	Type B
Gemma + GCG	33.3%	0.27	0.79	0.83	[0.34, 1.41]	Type B
Gemma + MCM	20.0%	0.09	0.22	0.24	[0.14, 0.31]	Type B

M.1 Inter-Judge Agreement

Table 25: Inter-judge agreement analysis ($n = 660$ harmful samples across all conditions).

Metric	Value
Overall agreement	90.0%
Both Safe (LG=0, HB=0)	56.2%
Both Unsafe (LG=1, HB=1)	33.8%
LG stricter (LG=0, HB=1)	1.3%
HB stricter (LG=1, HB=0)	8.7%

Interpretation. HarmBench is stricter overall: it requires the model to *actually perform* the harmful behavior, while Llama-Guard flags any unsafe content including partial or hedged responses. This explains the 8.7% of cases where Llama-Guard marks success but HarmBench marks failure.

M.2 Per-Condition ASR Comparison

Table 26: ASR comparison across judges.

Condition	LG ASR	HB ASR	ΔASR
Llama + MCM	8.3%	1.7%	+6.6%
Llama + GCG	15.0%	10.0%	+5.0%
Llama + DeepInc.	56.7%	30.0%	+26.7%
Mistral + MCM	5.0%	6.7%	-1.7%
Mistral + GCG	56.7%	58.3%	-1.6%
Mistral + DeepInc.	76.7%	68.3%	+8.4%
Qwen + MCM	51.7%	26.7%	+25.0%
Qwen + GCG	81.7%	73.3%	+8.4%
Qwen + DeepInc.	55.0%	45.0%	+10.0%
Gemma + MCM	20.0%	18.3%	+1.7%
Gemma + GCG	33.3%	33.3%	$\pm 0.0\%$

M.3 Type A/B Stability Across Judges

Classification stability. 9/10 comparable conditions (90%) maintain the same Type A/B classification across judges. The Llama+MCM discrepancy arises because HarmBench’s stricter labeling yields only 1 success (< 3 threshold for RP calculation). The sole classification mismatch (Mistral+MCM) reflects marginal RP differences. AUC values vary by < 0.05 across judges, confirming diagnostic robustness to judge choice.

Table 27: Type A/B classification comparison across judges.

Condition	Llama-Guard	HarmBench	Match
Llama + GCG	Type B	Type B	✓
Llama + MCM	Type A	N/A [†]	—
Llama + DeepInc.	Type B	Type B	✓
Mistral + GCG	Type A (inv)	Type A (inv)	✓
Mistral + MCM	Type B	Type A	×
Mistral + DeepInc.	Type A (inv)	Type A (inv)	✓
Qwen + MCM	Type A	Type A	✓
Qwen + GCG	Undetected	Undetected	✓
Qwen + DeepInc.	Type B	Type B	✓
Gemma + GCG	Type B	Type B	✓
Gemma + MCM	Type B	Type B	✓

[†]HarmBench: only 1 success (insufficient for RP calculation).

N Stochastic Decoding Analysis

N.1 Experimental Setup

We evaluate stochastic decoding robustness across temperature ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$), generating 10 independent trials per prompt across 4 decoy conditions, yielding 28,800 total trials.

N.2 Response Diversity

Table 28: Response diversity under stochastic decoding.

Temperature	Unique Response %	Groups >50% diverse
$T = 0.3$	54.2%	72.4%
$T = 0.7$	82.1%	91.7%
$T = 1.0$	92.2%	97.3%

N.3 Variance Decomposition

For pre-generation metrics (turn_start_S), within-prompt variance is **exactly zero** across all conditions (ICC = 1.0). This confirms S_0 is temperature-invariant by design—computed before any token sampling occurs. For generation-time metrics (step_S_mean), within-prompt variance is non-zero but small relative to between-prompt variance (ICC > 0.85 across conditions).

O Hidden-State Baseline Comparison

O.1 Methods

B1 (Refusal Direction): Following Ardit et al. [2024], we compute a refusal direction $\mathbf{d}$ using difference-of-means between refusal and compliance responses, then project hidden states at layer 16.

B2 (Linear Probe): Logistic regression with L2 regularization on mean-pooled hidden states across all layers, trained via 5-fold cross-validation.

O.2 Results

Key findings. Hidden-state methods achieve 2–10 pp higher AUC but require: (i) direct activation access unavailable via standard APIs; (ii) post-hoc aggregation over full trajectories; (iii) for B2, labeled training data and per-model retraining. Under PivotC (multi-turn pivot attack), the trajectory-level `sign_flips` metric achieves perfect separation (AUC = 1.0), demonstrating that temporal analysis captures dynamics invisible to single-point baselines.

Table 29: Hidden-state vs. logit-based diagnostic comparison (AUC, $n = 240$).

Family	Method	Base	DecoyA	DecoyB	PivotC
Hidden	B1 (direction)	0.974	0.983	0.974	0.974
Hidden	B2 (probe)	0.998	0.974	1.000	0.998
Logit	step_S_mean	0.913	0.893	0.904	0.998
Logit	turn_start_S	0.959	0.980	0.922	0.978
Logit	sign_flips	0.899	0.944	0.883	1.000

P Undetected Classification: Diagnostic Insensitivity vs. True Absence

The “Undetected” classification for Qwen+GCG ($r = 0.01$, 95% CI [0.00, 0.03]) admits two interpretations:

1. **True absence of safety engagement:** Qwen’s safety mechanisms genuinely fail to activate in response to GCG-style perturbations.
2. **Diagnostic insensitivity:** Safety mechanisms engage through pathways not captured by our lexicon-based probe (e.g., non-lexical internal representations).

P.1 Evidence for True Absence

Several observations support interpretation (1):

Extreme ASR. Qwen+GCG achieves 81.7% ASR—the highest across all conditions—suggesting systematic safety failure rather than marginal vulnerability.

Consistent near-zero signal. Both $RP_A = 0.00$ and $RP_B = -0.01$ are near-zero, indicating successful attacks are indistinguishable from harmful baseline along *both* diagnostic axes. If safety mechanisms engaged through alternative pathways, we would expect at least one axis to show differentiation.

Cross-model patterns. On the same GCG attack, Llama and Mistral show strong signals ($r = 2.96$ and $r = 0.35$ respectively), suggesting GCG attacks *can* elicit detectable safety responses when mechanisms are properly engaged. Qwen’s uniquely flat response suggests model-specific failure rather than attack-specific diagnostic blindness.

Entropy cross-validation. Generation-time entropy—a non-lexical metric—also shows no differentiation for Qwen+GCG ($|d| = 0.077$), while detecting signal on other conditions (Section P.4). This convergent evidence from independent metrics provides the strongest support for true safety mechanism absence.

P.2 Threshold Sensitivity Analysis

The “Undetected” threshold ($r < 0.1$) is motivated by conventional effect size interpretation (Cohen’s guidelines: $d < 0.2$ is “negligible”). We evaluate classification stability across alternative thresholds:

Table 30: Classification stability under varying r thresholds.

Condition	r	$r < 0.05$	$r < 0.10$	$r < 0.15$	$r < 0.20$
Qwen + GCG	0.01	Undet.	Undet.	Undet.	Undet.
Mistral + GCG	0.35	—	—	—	—
Qwen + MCM	0.58	—	—	—	—
Llama + MCM	1.12	—	—	—	—
Llama + GCG	2.96	—	—	—	—

Key findings. Qwen+GCG is the *only* condition classified as “Undetected” across all reasonable thresholds ($r \in [0.05, 0.20]$). The next-lowest r value (Mistral+GCG, $r = 0.35$) is $35\times$ larger than Qwen+GCG ($r = 0.01$), providing a wide separation margin. Even under the most conservative threshold ($r < 0.05$), Qwen+GCG remains classified as Undetected (CI upper bound = 0.03). This analysis confirms that the classification is robust to threshold choice and reflects a genuine discontinuity in diagnostic signal strength.

P.3 Validation Approach

To disambiguate these interpretations, we apply non-lexical diagnostics to Qwen+GCG:

Entropy validation (completed). Section P.4 reports that generation-time entropy shows $|d| = 0.077$, confirming lexicon-based findings with an independent metric. This supports interpretation (1): true safety mechanism absence.

Hidden-state validation (future work). Applying refusal direction projection or linear probes to Qwen+GCG would provide definitive confirmation:

- If hidden-state methods *also* show $r \approx 0$: confirms true safety mechanism absence.
- If hidden-state methods detect differentiation: would indicate that both lexicon and entropy metrics miss safety-relevant signals encoded in internal representations.

Our Llama hidden-state analysis (Section O) demonstrates that refusal direction projection achieves near-ceiling AUC (0.97–0.99), providing a strong baseline for such cross-validation.

P.4 Entropy Cross-Validation

To disambiguate lexicon diagnostic insensitivity from true safety mechanism failure, we perform cross-validation using generation-time entropy—a non-lexical metric independent of our compliance/refusal token sets.

Method. For each sample, we compute step-wise entropy $H_t = -\sum_v p_t(v) \log p_t(v)$ from softmax distributions during generation, then aggregate as $\bar{H}_{\text{gen}} = \frac{1}{T} \sum_{t=1}^T H_t$. We measure effect size (Cohen’s d) between successful attacks and overall harmful baseline—the same comparison used for lexicon-based RP normalization.

Table 31: Entropy cross-validation: Success vs. Overall Harmful.

Condition	ASR	Entropy $ d $	Lexicon $ d $	Interpretation
Qwen + GCG	81.7%	0.077	0.005–0.011	Undetected
Llama + MCM	10.0%	0.103–0.416	0.217–0.256	Weak–Medium

Results.

Interpretation. For Qwen+GCG, entropy-based differentiation yields $|d| = 0.077$ —below the $|d| < 0.1$ threshold for “no effect.” This matches the lexicon-based result ($|d| = 0.005\text{--}0.011$) and **supports interpretation (1): true safety mechanism failure**. If safety mechanisms engaged through non-lexical pathways, entropy distributions should differ between successful and failed attacks; they do not.

For contrast, Llama+MCM shows $|d| = 0.103\text{--}0.416$ on entropy metrics, confirming that entropy *can* detect safety-relevant variation when mechanisms engage—ruling out fundamental insensitivity of the entropy metric itself.

P.5 Implications for Defense

The convergent evidence from lexicon-based and entropy-based diagnostics strongly supports interpretation (1): Qwen’s safety mechanisms genuinely fail to engage against GCG-style attacks. This

indicates a fundamental safety gap in Qwen’s alignment, likely requiring architectural intervention or adversarial fine-tuning rather than detection-based defenses. The contrast with Llama and Mistral—which show detectable safety engagement under the same attack—suggests that GCG robustness varies substantially across model families.